Tutorial-10 Date:- 01/04/2026

Dataset:- (perfect classification)

Sample	Feature1 (Age)	Feature2 (Income)	Feature3 (Experience)	Target (Job offer)
S1	24	35	1	NO
S2	30	50	3	Yes
S3	45	40	10	Yes
S4	35	30	7	No
S5	28	55	2	Yes

Target Encoding

Yes = +1

No = -1.

Questions:-

1. Assign initial Weights to all samples.
2. construct a decision stump using Feature2 (Income).
3. Identify Misclassified samples.
4. Compute the Weighted classification error.
5. Calculate the classifier Weight.

1. Initial Weights:

$W_i = \frac{1}{N} = \frac{1}{5} = 0.2$. (use for all samples).

Here, N=5

2. Construct Decision stump.

Decision stump (Weak Learner) on Feature2 (Income).

Income $< 38 \rightarrow \times$

$\geq 38 \rightarrow \checkmark$

$h_t(x_i)$

$h_1 \rightarrow$ first decision stump.

Target (y)	predict ($h_1(x_i)$)
-1	-1
$+1$	$+1$
$+1$	$+1$
-1	-1
$+1$	$+1$

Target & predict Same $= 0$

different $= 1$

$\rightarrow$ It is a perfect classification.

3. It is a perfect classification. There is no misclassification Examples.

4.

Weighted Error:-

$$\varepsilon_1 = \sum_{i=1}^{N} w_i I(y_i \neq h_1(x_i))$$

Input

$$= [0.2 \times 0 + 0.2 \times 0 + \cdots] = 0$$

Weighted Error is '0'.

5.

Classifier or Model Error:-

$t=1$ $\quad \alpha_1 = \frac{1}{2} \ln\left(\frac{1-\varepsilon_1}{\varepsilon_1}\right)$

$$= \frac{1}{2} \ln\left(\frac{1-0}{0}\right)$$

$$\alpha_1 = \infty \text{ (undefined)}$$

6. Weight update:-

$$w_i^{new} \rightarrow w_i^{old} e^{-y_i \alpha_i h_i(x_i)}$$

for 1

$$w_1^{new} \rightarrow w_1^{old} e^{-y_1 \alpha_1 h_1(x_1)}$$

$$w_1^{new} \rightarrow 0.2 \times e^{-(-1) \cdot \infty (-1)}$$

$$w_1 \rightarrow 0$$

$$\boxed{e^{-\infty} = 0}$$

for 2, 3, 4, 5

$$w_2^{new} \rightarrow 0, \; w_3^{new} \rightarrow 0, \; w_4^{new} \rightarrow 0, \; w_5^{new} \rightarrow 0.$$

for each sample it is '0'. Here, the model stops training. Because perfectly classified.

$w_1 = 0, \; w_2 = 0, \; w_3 = 0, \; w_4 = 0, \; w_5 = 0.$

7) Final predict.

$$H(x) = \text{sign}\left(\sum_{t=1}^{T} \alpha_t h_t(x_i)\right)$$

$$= \text{sign}(\alpha_1 h_1(x_1))$$

$$= \text{sign}(\infty)$$

$=$ ↳ positive or Negative.

Missclassification:—

Sample	Feature1 (Age)	Feature2 (Income)	Feature3 (Experience)	Target (Job offer)
S_1	24	35	1	No
S_2	30	50	3	Yes
S_3	45	40	10	Yes
S_4	35	30	7	No
S_5	28	55	2	No

1. Calculate Initial Weights

$$w_i = \frac{1}{N} \quad \text{where } N=5$$

$$= \frac{1}{5} = 0.2.$$

2. Construct Decision stump. Based on Feature2 (Income)

Income $< 38 \rightarrow \times$
$\geq 38 \rightarrow \checkmark$

Target (y)	predict ($h_1(x_i)$)
-1	-1
+1	+1
+1	+1
-1	-1
-1	+1

$\longrightarrow$ Miss classification.

3. It is a Misseclassification. 5 is the Misclassified Sample.

4. Weighted Error:-

$$\varepsilon_1 = \sum_{i=1}^{N} w_i \, I(y_i \neq h_2(x_i))$$

$$= [0.2\times0 + 0.2\times0 + 0.2\times0 + 0.2\times0 + 0.2\times1]$$

$$= 0.2.$$

5. Classifier or Model Error:-

$$\alpha_1 = \frac{1}{2}\ln\left(\frac{1-\varepsilon_1}{\varepsilon_1}\right)$$

$$= \frac{1}{2}\ln\left(\frac{1-0.2}{0.2}\right)$$

$$\alpha_1 = 0.693$$

6. Weight updates

$$w_i^{new} \rightarrow w_i^{old} \cdot e^{-y_i \alpha_i h_i(x_i)}$$

$$w_1^{new} \rightarrow w_1^{old} \cdot e^{-y_1 \alpha_1 h_1(x_1)}$$

$$= 0.2 \cdot e^{-(-1)0.693(-1)}$$

$$= 0.1$$

$$w_2^{new} \rightarrow w_2^{old} \cdot e^{-y_2 \alpha_2 h_2(x_2)}$$

$$= 0.2 \cdot e^{-(1)0.693(1)}$$

$$= 0.1$$

$$w_3^{new} \rightarrow w_3^{old} \cdot e^{-y_3 \alpha_3 h_3(x_3)}$$

$$= 0.2 \cdot e^{-(1)0.693(1)}$$

$$= 0.1$$

$$w_4^{new} \rightarrow w_4^{old} \cdot e^{-y_4 \alpha_4 h_4(x_4)}$$

$$= 0.2 \cdot e^{-(-1)0.693(-1)}$$

$= 0.1$

$$W_5^{new} \rightarrow W_5^{old} \cdot e^{-y_5 \alpha_5 h_5(x_5)}$$

$$= 0.2 \cdot e^{-(-1)0.693(1)}$$

$$= 0.4$$

7. Normalized Weight.

$$W_1^N = \frac{W_1^{new}}{\sum_{i=1}^{N} W_i^{New}} = \frac{0.1}{0.8} = 0.125$$

$$W_2^N = \frac{W_2^{new}}{\sum_{i=1}^{N} W_i^{New}} = \frac{0.1}{0.8} = 0.125$$

$$W^N = \frac{W_3^{New}}{\sum_{i=1}^{N} W_i^{New}} = \frac{0.1}{0.8} = 0.125$$

$$W^N = \frac{W_4^{New}}{\sum_{i=1}^{N} W_i^{New}} = \frac{0.1}{0.8} = 0.125$$

$$W^N = \frac{W_5^{New}}{\sum_{i=1}^{N} W_i^{New}} = \frac{0.4}{0.8} = 0.5$$

8. Final predict

$$H(X) = Sign(\alpha_1 h_1 + \alpha_2 h_2 + \cdots \alpha_5 h_5)$$

$h_1(x_2) = +1$

$$= Sign\left[(0.693 \times 1) + \cdots + \cdots\right]$$

If two Misclassification samples there. We Should take Maximum Normalized Weight. for h.

10.04.21